COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 25 — Agile Project Management (Scrum & Sprint Planning)

Section 14: Classroom Activity — Plan a Two-Week Sprint

Sprint planning simulation for the first 2-week execution sprint (Sprint 1) of the College Event Management System.

1. Product Goal & Sprint Goal

- **Product Goal:** Provide an intuitive, rock-solid web platform for students to browse and register for all campus festival events with zero overbooking.
- **Sprint 1 Goal:** Deliver a demonstrable end-to-end event discovery and atomic registration workflow with live database seat deduction.

2. Sprint Backlog & Story Point Estimation

Story ID	User Story Description	Story Points	Sprint 1 Inclusion
US-01	As a student, I want to browse active events by category, so I can discover workshops.	3 SP	Committed
US-02	As a student, I want to register for an event, so that my seat is reserved.	5 SP	Committed
US-03	As a student, I want to view my confirmed bookings, so that I have proof of entry.	3 SP	Committed
US-04	As a coordinator, I want to define event seat capacity, so venues do not overfill.	3 SP	Committed
US-05	As a student, I want to cancel a registration, so I can free my seat for others.	5 SP	Committed
US-06	As an admin, I want to generate QR code check-in badges.	8 SP	Deferred (Sprint 2)
US-07	As an organizer, I want to export CSV attendance reports.	5 SP	Deferred (Sprint 2)

US-08	As an executive, I want an analytics dashboard of student demographics.	8 SP	Deferred (Sprint 3)

Team Velocity: Assumed capacity of 4 developers = 20 Story Points for Sprint 1. Total committed: 19 SP.

3. User Story with Testable Acceptance Criteria

- **Story US-02:** "As a student, I want to register for an open event, so that I secure attendance."
- **Acceptance Criteria 1:** Given an event with available seats, when the student clicks "Register", then a booking record is created, available seats decrement by 1, and status shows "CONFIRMED".
- **Acceptance Criteria 2:** Given an event at maximum capacity, when a student attempts registration, then the submission is rejected with "Event Full" and seat count does not change.
- **Acceptance Criteria 3:** Given a student already registered for Event X, when they attempt to register for Event X again, then system blocks the request with "Duplicate Registration Error".

4. Definition of Done (DoD)

- **Definition of Done:** (1) Code compiles with zero warnings; (2) Unit tests pass with >= 85% branch coverage; (3) Peer code review approved by at least 1 senior developer; (4) Deployed and verified on staging environment; (5) Product Owner accepts all acceptance criteria.

Section 17: Homework — Scrum Backlog & Artifact Analysis

Scrum framework artifact differentiation and backlog management for campus systems.

- **1. Ten-Item Product Backlog for Library System:** Ranked Backlog: (1) Book Catalog Search; (2) Patron Barcode Authentication; (3) Book Issue & Loan Ledger; (4) Book Return Check-in; (5) Automated Overdue Fine Computation; (6) Email Due-Date Reminders; (7) Online Book Reservation Hold; (8) Inter-Library Loan Request; (9) Digital Fine Payment Gateway; (10) Annual Loss & Weeding Report.
- **2. Artifact Differentiation (Product Backlog vs Sprint Backlog vs Increment):** Product Backlog: Single, living, prioritized source of requirements for the entire product lifecycle; managed by Product Owner. Sprint Backlog: Subset of Product Backlog items selected by Development Team for execution during current Sprint, broken into granular engineering tasks. Increment: The sum of all Product Backlog items completed during a Sprint and all previous Sprints, meeting the Definition of Done and representing usable, potentially shippable software.
- **3. Backlog Reordering Reflection:** Feedback indicating that 40% of overdue fines are disputed due to holiday miscalculations would trigger an immediate reordering of the Product Backlog, pulling the "Configurable Academic Holiday Fine Calendar" story ahead of new features like "Inter-Library Loans".